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Highlights

- Ultrafast laser welding achieves sound joints in 45vol.% SiC_p/Al composites.
- In-situ nanoscale Al₄SiC₄ uniformly forms in the welding seam, free from Al₄C₃.
- A bending strength of 321 MPa is achieved, 60% of the base material's value.

Abstract

Outline

Highlights

Abstract

Keywords

1. Introduction

2. Materials and methods

3. Result and discussion

4. Conclusion

CRediT authorship contribution statement

Ceramic-reinforced aluminum matrix composites possess exceptional properties, which makes them promising candidates for a wide range of applications. However, the brittle Al_4C_3 phases, which can severely deteriorate the properties of the joint, are often formed in the joint obtained using conventional fusion welding techniques. In this study, an ultrafast laser welding approach is adopted for the first time to join aluminum matrix composites reinforced with 45vol.% SiC particles. The achieved joints appear to be sound, free from macroscopic defects, and the four-point bending strength of the joint can reach 321 MPa. The effects of laser power and welding speed on the microstructure and mechanical properties are systematically examined. Comprehensive characterization reveals that, unlike conventional laser welding, no brittle and hydrolyzable Al_4C_3 phase is detected in the welding seam. Instead, the ultrafast laser irradiation promotes the decomposition of SiC particles, which subsequently react with molten aluminum to form nanoscale Al_4SiC_4 phases. These precipitates are uniformly dispersed throughout the matrix, thus enhancing the joint performance.

Keywords

Ultrafast laser welding; SiC_p/Al composites; Microstructure; Mechanical properties

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1. Introduction

Ceramic-reinforced aluminum matrix composites are a class of advanced materials fabricated by incorporating ceramic particles—such as SiC, Al_2O_3 , B_4C , and TiC—into an aluminum alloy matrix. Compared with conventional aluminum alloys, these composites exhibit superior specific strength, specific stiffness, wear resistance, corrosion resistance, thermal stability, and fatigue performance [1], [2], [3]. These materials possess a unique combination of properties, which is unattainable with monolithic aluminum or ceramics

Declaration of competing interest

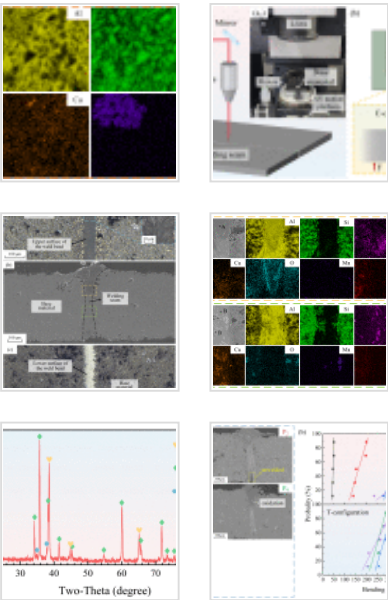
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[4], [5]. With the increasing heat dissipation demands of electronic systems and high-power chips, the development of novel materials that combine efficient thermal management capabilities with high reliability has drawn significant attention. SiC reinforced aluminum matrix composites with high volume fraction have attracted great interest [6]. These composites offer excellent properties, including high thermal conductivity, high strength, and high hardness, enabling them to meet the critical simultaneous requirements for heat dissipation and structural rigidity in advanced applications [7], [8].

However, machining ceramic-reinforced aluminum matrix composites remains difficult, and near-net-shape forming processes are often inadequate for producing complex components [9]. These challenges still remain unresolved, thereby limiting the full potential of these high-performance composites in advanced applications [10]. Consequently, welding technologies are essential for assembling simpler parts into intricate structures and for performing repairs. Nevertheless, ceramic-reinforced aluminum matrix composites exhibit poor weldability. This is due to the significant physicochemical differences between the ceramic reinforcement and the metal matrix, and the high chemical reactivity of aluminum. These factors lead to the formation of brittle intermetallic compounds in the welding seam. These challenges severely restrict the widespread deployment of ceramic-reinforced aluminum matrix composites.

Conventional joining techniques for ceramic-reinforced aluminum matrix composites, such as conventional laser welding, brazing, and friction stir welding, each face certain limitations. Conventional fusion welding usually leads to the formation of brittle phases at the Al/ceramic interface and consequent joint degradation [11], [12]. Specifically, at the interface, SiC reacts with molten Al according to the following reaction: $4\text{Al (l)} + 3\text{SiC (s)} \rightarrow \text{Al}_4\text{C}_3 \text{ (s)} + 3\text{Si (s)}$. The Al₄C₃ nucleates preferentially on the SiC particle surfaces, a typical case of heterogeneous nucleation. Due to the anisotropy in growth rates along different crystallographic directions, the Al₄C₃ tends to form needle-like or plate-like morphologies [13], [14]. The intrinsically brittle Al₄C₃ phases act as sites for crack initiation and propagation. Moreover, the susceptibility of Al₄C₃ to hydrolysis further

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Supplementary figures

degrades the mechanical properties of the joint [15], [16]. A key challenge in brazing is selecting a filler material that effectively wets both the ceramic reinforcement and the aluminum matrix [17], [18]. Furthermore, the inherent limitations of available fillers often lead to restricted joint strength [19], [20]. Friction stir welding is hindered by severe tool wear caused by the high hardness of the reinforcement [21], [22]. Since this wear cannot be fully eliminated even with protective coatings, this technique has limited applicability for highly reinforced composites or long joints [23].

When such materials are applied to precision and complex components, high demands are placed on joining accuracy. However, conventional welding methods struggle to achieve high-precision joints, which has become a major technical bottleneck hindering their engineering application. Therefore, it is urgent to develop a new joining method to enable effective bonding between them.

Ultrafast lasers, characterized by ultrashort pulse widths on the picosecond or femtosecond scale, generate extremely high peak power densities and low heat input. In 2005, Takayuki et al. [24] at Osaka University employed a femtosecond laser to weld silica glass, thereby pioneering the field of ultrafast laser welding. Further studies have revealed that the microstructure and mechanical properties of joints fabricated via ultrafast laser welding are influenced by numerous factors, including the intrinsic material properties, laser parameters, and energy density [25], [26]. Ultrafast laser irradiation induces nonlinear absorption in ceramics and glasses, in contrast to linear absorption in metallic materials [27], [28]. These absorption effects lead to subsequent processes, including melting and ablation, and bonding is ultimately achieved upon cooling [29], [30]. This technique is characterized by minimal heat-affected zone formation, negligible thermal damage, precise energy control, and high processing flexibility [31]. Such advantages make ultrafast laser welding a promising technique for joining ceramic-reinforced aluminum matrix composites.

Indeed, current research on ultrafast laser welding primarily focuses on material pairs including glass-to-glass, glass-to-metal, glass-to-ceramic, glass-to-semiconductor,

ceramic-to-ceramic, and ceramic-to-metal. However, studies on the welding of ceramic-reinforced metal matrix composites remain scarce.

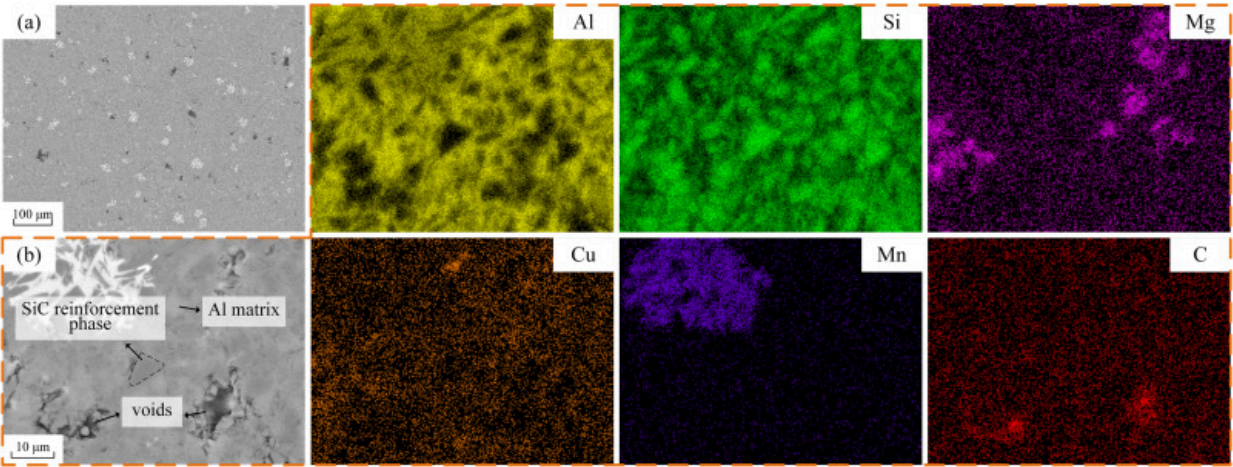
Thus, investigating the process and underlying mechanisms of ultrafast laser welding for SiC-reinforced aluminum matrix composites is of considerable importance. In this study, sound welded joints of 45vol.% SiC_p/Al composites were achieved via ultrafast laser welding. Under optimized parameters, the joint exhibited a four-point bending strength of 321 MPa with no evident defects. Furthermore, the underlying welding mechanism was elucidated.

2. Materials and methods

SiC_p/Al composites with 45vol.% SiC particles, produced by a pressure infiltration method, were used as the base material. The preparation process included the following steps: First, SiC particles were preheated and placed into a mold to form a preform. Then, aluminum alloy was melted under an argon atmosphere and injected into the mold. During infiltration, pressure was applied, and the composite was subsequently solidified through cooling in air.

As shown in the macroscopic morphology and corresponding EDS mapping results ([Fig. 1](#)), the dark gray particles represent the SiC reinforcement phase, while the light gray matrix corresponds to the aluminum alloy. The matrix contained small amounts of Cu and Mg as alloying elements, with isolated bright white regions indicating Mn-rich phases. The contents of these elements were quantified by inductively coupled plasma mass spectrometry (ICP-MS, PerkinElmer, USA) and are listed in [Table 1](#). The microstructure also exhibits some voids. In the case of the 45vol.% SiC/Al composite, which contains a high volume fraction of reinforcement, porosity is a prevalent defect. The formation of these voids can be attributed to two main reasons. Poor wettability between molten aluminum and SiC particles, along with the high melt viscosity caused by the high particle content, prevents the complete infiltration of interparticle spaces and leads to inadequate densification. Additionally, the agglomeration of SiC particles and the

significant thermal expansion mismatch with the aluminum matrix lead to the formation of stress concentrations and voids [32].



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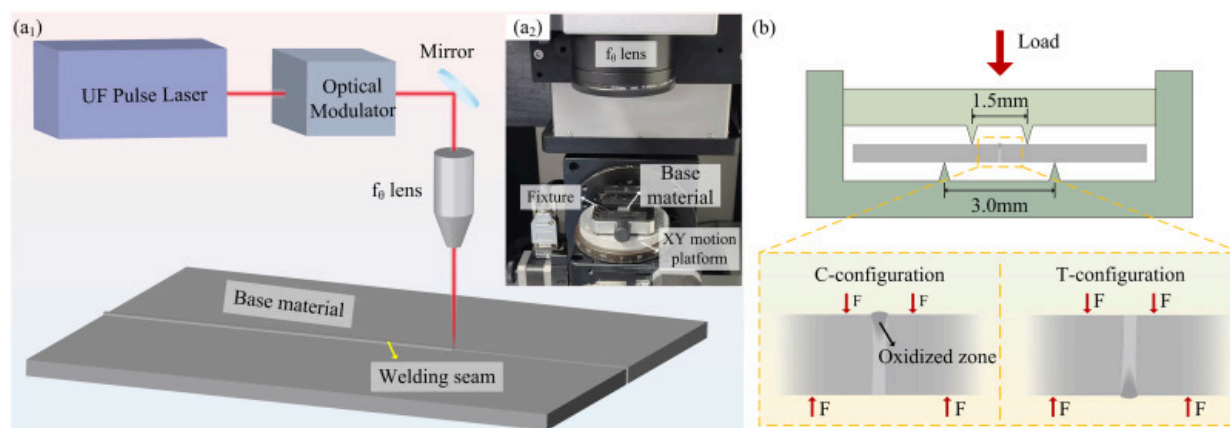
Fig. 1. SEM image (a) and corresponding elemental mapping (b) of the base material.

Table 1. Content of alloying elements in the matrix.

Element	Cu	Mg	Mn
Content (wt.%)	1.67	0.61	0.36

The base material was cut into specimens with the size of 12mm×4mm×0.5mm by diamond wire cutting. The welding surface (12mm×0.5mm) was ground with 400#, 800#, 1000#, 1500#, 2000#, and 3000# SiC sandpapers, and finally polished with a 0.05μm SiO₂ suspension. Finally, the samples were ultrasonically cleaned in absolute ethanol for 5 min.

The welding system primarily consisted of an infrared ultrashort pulse laser, an XY motion platform, and a centralized control system. The laser source operated at a wavelength of 1030nm, a repetition rate of 1 MHz, with a pulse width of 300fs. The laser beam was focused on the workpiece surface with zero defocusing distance. No shielding gas was applied during the welding process. The schematic diagram of the ultrafast laser welding is illustrated in Fig. 2(a₁), with a corresponding actual fabricated picture shown in Fig. 2(a₂). During fixturing, the workpieces were in close contact at the weld start point, with a pre-set gap of approximately 15μm at the weld end to compensate for welding-induced deformation.



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Fig. 2. (a₁) Schematic of the ultrafast laser welding; (a₂) photograph of the corresponding laboratory fabricated pictures; (b) schematic of the four-point bending test configuration.

Microstructural characterization was performed using a Gemini-560 field-emission scanning electron microscope (SEM, Carl Zeiss, Germany). Phase constituents of the welding seam were identified by X-ray diffraction (XRD, D8 Advanced, Bruker, Germany). For Transmission Electron Microscope (TEM) analysis, specimens were prepared from targeted regions using a focused ion beam (FIB) system (TESCAN AMBER). These specimens were then examined by bright-field (BF) imaging, scanning transmission

electron microscopy (STEM), high-resolution TEM (HRTEM), and selected-area electron diffraction (SAED) using a Talos F200X in-situ multifunctional TEM (Thermo Fisher Scientific, USA).

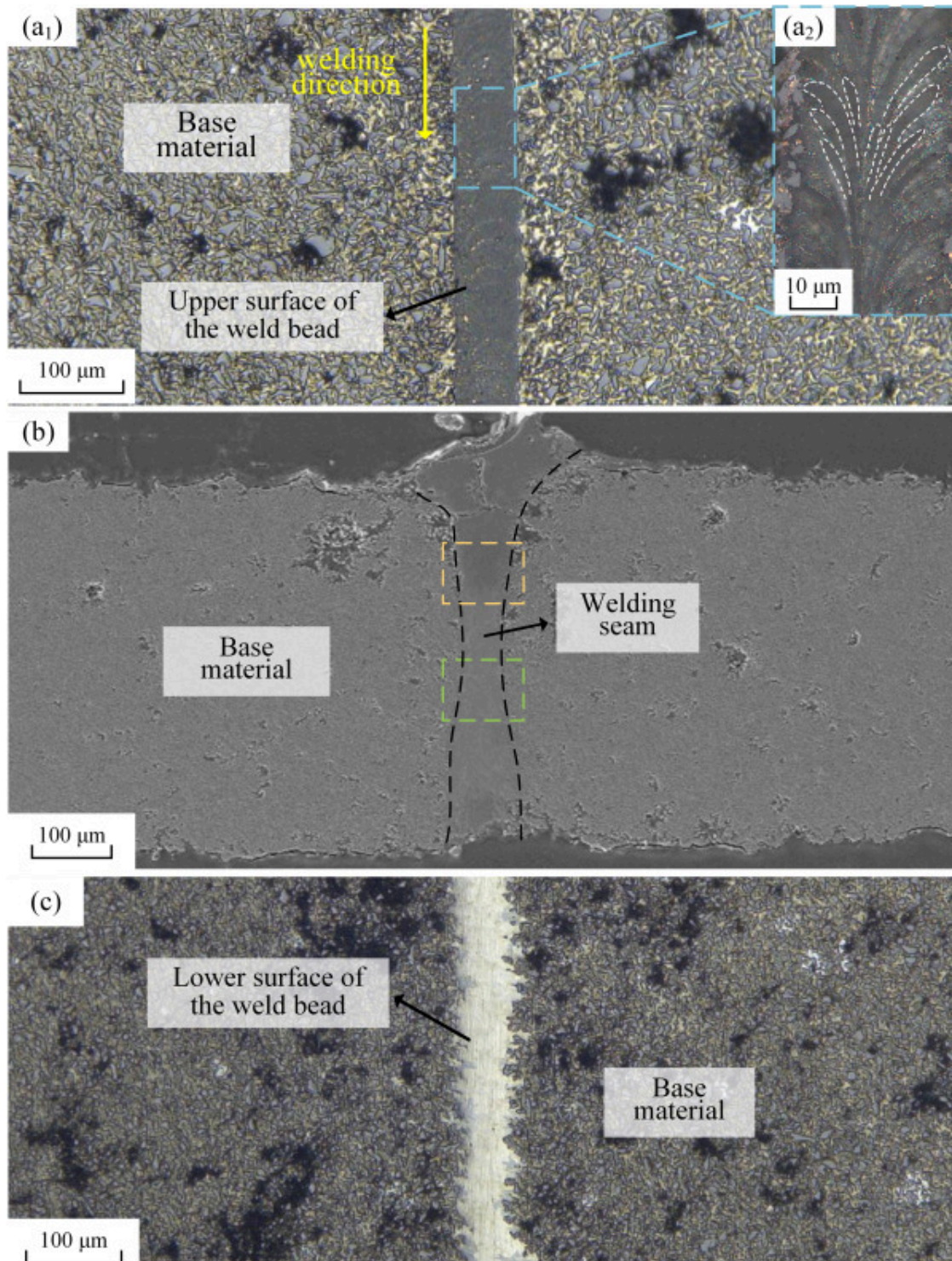
The 45 vol.% SiC_p/Al composite used in this study exhibits pronounced brittleness and low ductility characteristics due to its high reinforcement fraction. For such materials, conventional tensile tests are often unsuitable since they typically produce highly scattered and unreliable strength data. To overcome this limitation and obtain a statistically robust strength evaluation, we employed the four-point bending test method, as it is particularly suited for brittle materials. The four-point bending strength of the welded joints was measured using a universal testing machine equipped with a four-point bending fixture. Owing to inherent process instabilities during welding initiation and termination, the first and last 2 mm of each joint were removed to ensure the reliability of mechanical property data. The bending specimens had dimensions of 8 mm × 8 mm × 0.5 mm. The fixture had outer and inner spans of 3 mm and 1.5 mm, respectively. Tests were conducted at a constant crosshead speed of 0.5 mm/min. For each welding parameter set, a minimum of five specimens were tested.

3. Result and discussion

3.1. Typical microstructure of the joint

Fig. 3 displays an image of a representative joint fabricated with a laser power of 19.78 W and a welding speed of 0.05 mm/s. These conditions resulted in full penetration welding, producing a high-quality joint. Columnar grains observed on the upper bead surface indicate that this region experienced melting and directional solidification. Furthermore, the grains grew along the welding direction. This preferential growth confirms the occurrence of directional solidification. Due to complete melting and rapid solidification, the joint exhibits a refined microstructure and a density significantly exceeding that of the base material. The joint is free from defects such as voids, hydrolysis pits, or cracks, demonstrating excellent structural integrity. Weld reinforcement was observed, with a

weld width of approximately 60 μm . No obvious welding distortion was detected, which is attributable to the ultrashort pulse duration of the ultrafast laser, thereby restricting thermal input and the resultant material damage to negligible levels.

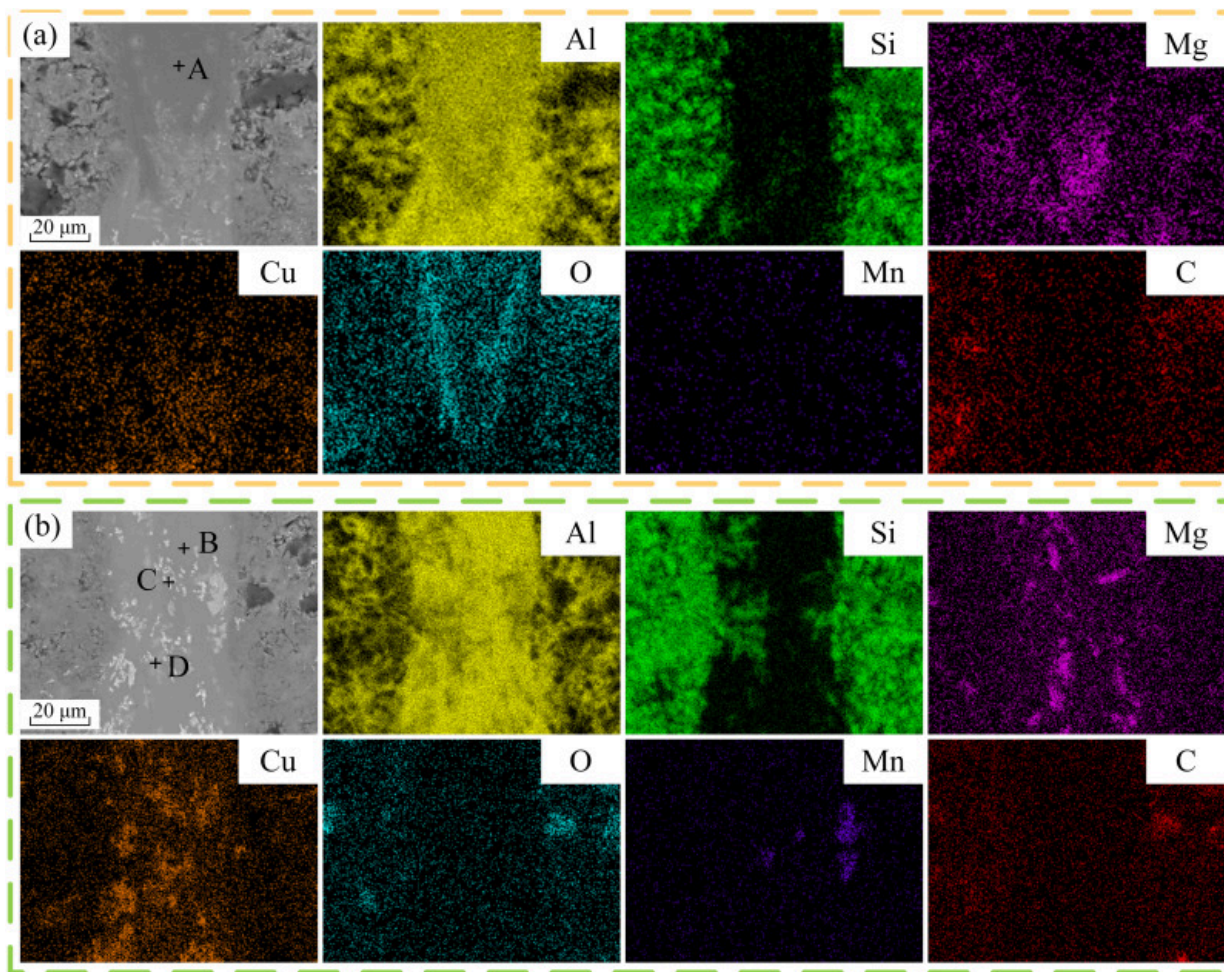


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Fig. 3. Microstructure of the weld bead: (a₁) overview of the upper surface; (a₂) higher-magnification view of the region marked in (a₁), revealing the presence of columnar crystals; (b) cross-sectional view; (c) lower surface morphology.

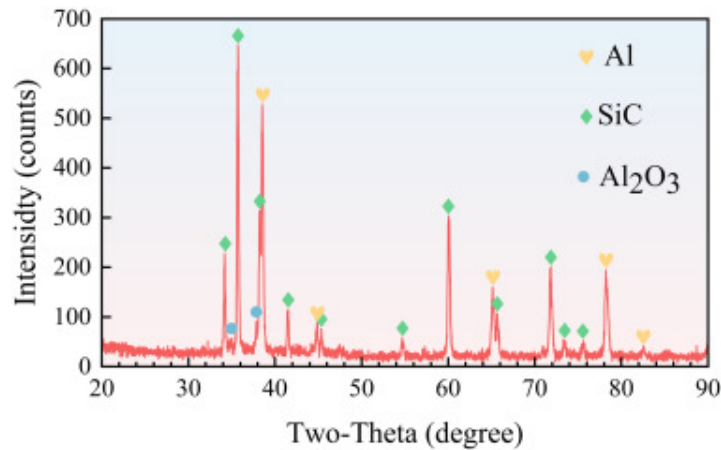
Higher-magnification SEM imaging, complemented by EDS analysis, reveals that the joint microstructure comprises four distinct zones with differing chemical compositions. These zones are marked A to D in Fig. 4. The dark gray Phase A, observed in the top welding seam region, is identified as Al₂O₃ via combined EDS and XRD analysis, as shown in Fig. 4, Fig. 5. The oxidized zone resulted from the high reactivity of aluminum under the elevated temperatures characteristic of the welding process. The light gray Phase B corresponds to Al matrix. Phases C and D, with contrast differing from the matrix, are attributed to the enrichment of Cu and of Si-Mg-Cu in the Al matrix, respectively. As shown in Table 1, the alloying element content in the aluminum matrix is very low. Consequently, compounds formed by the reaction of these elements with Al are difficult to detect by XRD. To address this, we extracted samples from Cu-enriched regions using FIB for further characterization. As presented in Fig.A.1, the corresponding SAED patterns confirm that PhaseC is a Cu-rich phase identified as Al₂Cu, which is consistent with the common strengthening phase found in aluminum alloys welded by conventional laser methods [33]. Based on typical phases observed in conventional laser welding, PhaseD is inferred to be AlCuMgSi [34], [35].The corresponding elemental compositions and the possible phases identified are summarized in Table 2.



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Fig. 4. Higher-magnification micrographs and corresponding elemental mapping of typical regions in the welding seam marked in Fig. 3(b): (a) top region; (b) center and bottom region.



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Fig. 5. XRD pattern of the joint.

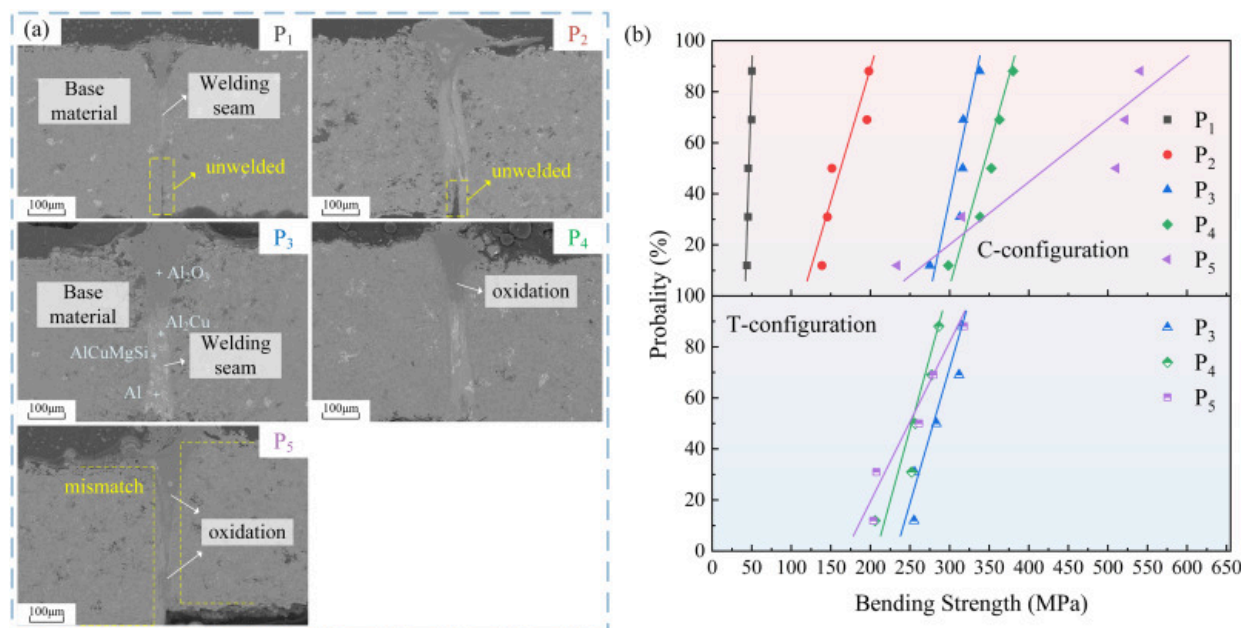
Table 2. EDS elemental composition (at. %) of phases marked in Fig. 4.

Position	Al	Si	C	O	Cu	Mg	Mn	Possible phase
A	36.83	0.02	17.51	45.55	0.03	0.05	0.01	Al ₂ O ₃
B	49.39	2.44	44.73	2.29	0.69	0.42	0.05	Al
C	50.12	0.98	37.81	2.11	8.55	0.39	0.05	θ-Al ₂ Cu
D	32.17	8.96	46.51	1.16	2.62	8.57	0.01	Q-AlCuMgSi

3.2. The effects of laser power and welding speed on microstructure and mechanical properties of joints

Laser power and welding speed are the primary parameters controlling heat input in ultrafast laser welding. Therefore, it is necessary to explore the effect of laser power and welding speed on the microstructure and mechanical properties of the joint.

Fig. 6(a) presents the cross-sectional microstructure of joints welded at a constant welding speed of 0.05 mm/s with laser power ranging from 13.85 to 25.03 W. With increasing laser power, both the weld width and penetration depth expanded progressively due to higher heat input, accompanied by more severe oxidation. Complete penetration was achieved when the laser power reached 17.79 W. However, when the power exceeded 25.03 W, the workpieces, although initially well-aligned, exhibited a significant mismatch toward the middle and end of the joint. Excessive heat input caused non-uniform thermal deformation during welding, resulting in a mismatch after solidification.



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Fig. 6. (a) Cross-sectional microstructure and (b) four-point bending strength, represented by Weibull distribution, of the joints produced at different laser powers and a constant welding speed of 0.05 mm/s, where P_1 , P_2 , P_3 , P_4 , and P_5 represent laser powers of 13.85 W, 17.79 W, 19.78 W, 21.62 W, and 25.03 W, respectively.

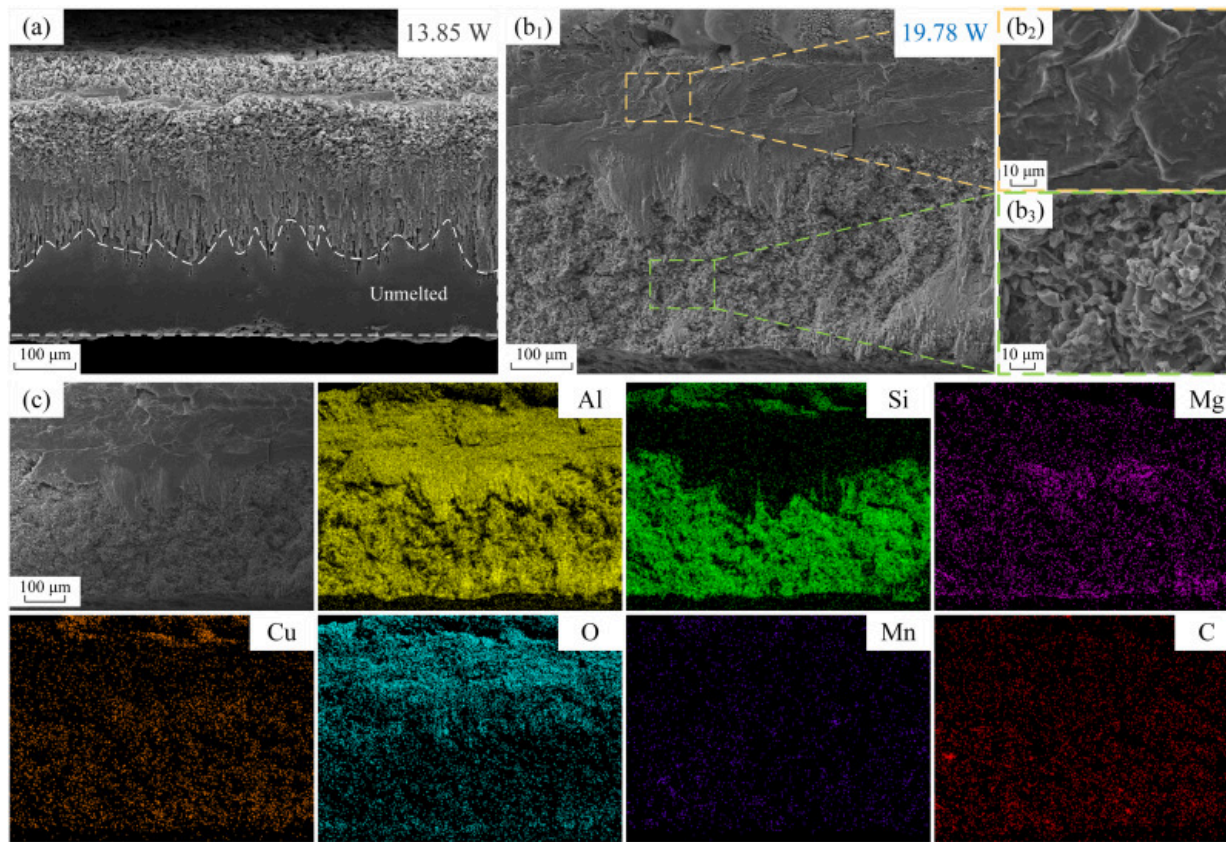
Considering potential property variations induced by oxidation-related microstructural heterogeneity, four-point bending tests were conducted in two distinct loading orientations (as illustrated in Fig. 2(b)): (1) C-configuration: The laser-irradiated surface (oxidized zone) was oriented upward, inducing compressive stress on the oxidized region during loading; (2) T-configuration: The specimen was inverted to position the laser-irradiated surface downward, thereby applying tensile stress to the oxidized zone. Owing to the brittle nature of the material, the bending strength data were analyzed using a two-parameter Weibull distribution to estimate the probability of failure and assess the reliability of the mechanical performance.

The four-point bending strength is depicted in Fig. 6(b), with the corresponding statistical parameters summarized in Table 3. Note that P_1 , P_2 , P_3 , P_4 , and P_5 represent laser powers of 13.85W, 17.79W, 19.78W, 21.62W, and 25.03W, respectively. Under C-configuration testing, joint strength exhibited a positive correlation with laser power. The lowest-power specimens exhibited critically low strength values below 100MPa. Based on combined analysis with SEM micrographs, this was attributed to incomplete penetration resulting from insufficient heat input. A laser power of 19.78W produced full penetration. The resulting joint exhibited a bending strength of 321MPa, which reaches approximately 60% of the base material's strength. At this power, oxidation was limited, resulting in a modest strength difference between the C and T configurations. Based on favorable microstructural characteristics, this parameter is considered optimal, with a corresponding calculated linear energy of 395.6J/mm. At higher power levels, oxidation intensified and its effect on strength became more evident. At 25.03W, the C-configuration specimens exhibited substantial scatter in strength (Weibull shape parameter $k=4.2$), which may be attributed to the combined effects of enhanced oxidation and increased thermal distortion.

Table 3. Weibull distribution parameters of the joint four-point bending strength as a function of laser power.

Parameter	P ₁ -C	P ₂ -C	P ₃ -C	P ₄ -C	P ₅ -C
Bending strength λ (MPa)	48.3	177.1	320.9	358.7	470.4
Shape parameter k	20.7	7.2	19.6	16.2	4.2
Parameter	–	–	P ₃ -T	P ₄ -T	P ₅ -T
Bending strength λ (MPa)	–	–	296.2	267.5	272.7
Shape parameter k	–	–	12.8	12.2	6.6

Fig. 7 compares the four-point bending fracture morphologies of specimens welded at laser powers of 13.85W and 19.78W. As can be seen from Fig. 7(a), when the laser power is 13.85W, the lower region of the fracture surface displays a smooth, planar morphology, indicating that the base material remained unmelted, which is consistent with the lack of penetration observed in the cross-sectional microstructures. In contrast, when the laser power was set at 19.78W (Fig. 7(b₁–b₃)), the joint achieved full penetration. The elemental distribution on the fracture surface (Fig. 7(c)) suggested that the fracture have occurred within or near the welding seam. Additionally, the overall morphology of the fracture surface exhibited characteristic features of brittle fracture.



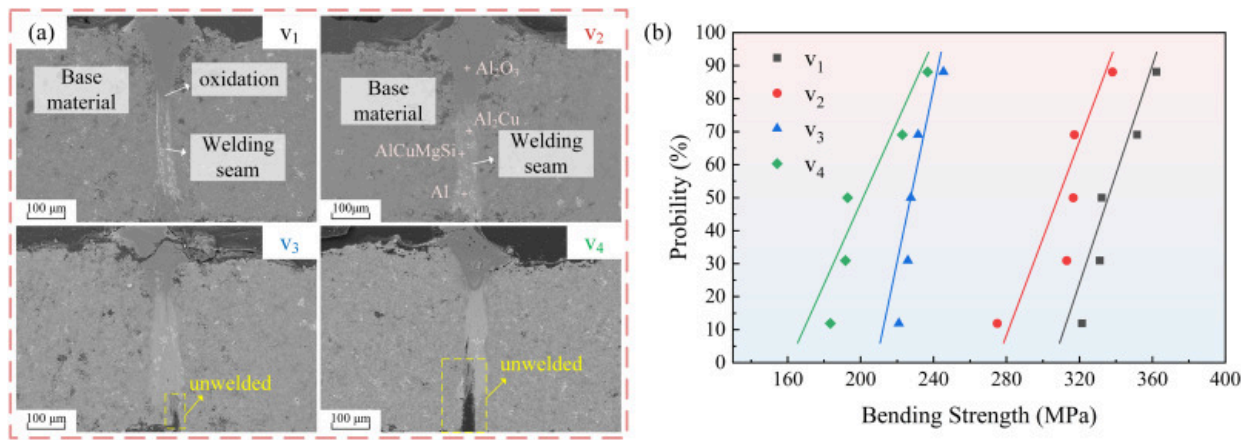
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Fig. 7. Fracture morphology after four-point bending test at 13.85W (a) and 19.78W (b₁)–(b₃), and elemental distribution of the fracture surface obtained at 19.78W (c).

The laser power was fixed at 19.78W, having been identified as optimal from the previous results. Under this condition, variations in welding speed resulted in distinct changes in the joint microstructure, as shown in Fig. 8(a). The corresponding Weibull distribution of the four-point bending strength of the joint is presented in Fig. 8(b), with detailed statistical data provided in Table 4. Here, v_1 through v_4 denote welding speeds of 0.03 mm/s, 0.05 mm/s, 0.07 mm/s, and 0.09 mm/s, respectively. An increase in welding

speed lowered the heat input, reducing oxidation phenomena but also causing a decline in joint strength. Complete penetration was not achieved when the welding speed exceeded 0.07 mm/s. At the maximum speed of 0.09 mm/s, insufficient heat input resulted in incomplete penetration, resulting in a bond only in the upper zone. The subsequent solidification shrinkage of this region, combined with the lack of mechanical constraint in the lower region, ultimately induced angular distortion.



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Fig. 8. (a) Cross-sectional microstructure and (b) four-point bending strength, represented by Weibull distribution, of the joints produced at different welding speeds and a constant laser power of 19.78 W, with v_1 , v_2 , v_3 , and v_4 denote welding speeds of 0.03 mm/s, 0.05 mm/s, 0.07 mm/s, and 0.09 mm/s, respectively.

Table 4. Weibull distribution parameters of the joint four-point bending strength as a function of welding speed.

Parameter	v_1 -C	v_2 -C	v_3 -C	v_4 -C
Bending strength λ (MPa)	347.1	320.9	234.6	215.2

Parameter	v_1-C	v_2-C	v_3-C	v_4-C
Shape parameter k	24.1	19.6	26.0	10.6

For joining high-volume-fraction (>40%) SiC/Al composites, brazing is the most widely used method, while fusion welding and friction stir welding are seldom reported. Table 5 summarizes the joint strengths achieved via brazing or soldering techniques in recent studies. In comparison, joints fabricated by ultrafast laser welding offer advantages in strength. Furthermore, ultrafast laser welding enables high-precision and micro-zone welding, which is difficult to achieve with brazing or soldering. Notably, some references did not report the base material strength or strength ratio relative to the base material. For the 55 vol.% SiC_p/Al joints, the strength ratio was calculated based on the shear strength value of 149 MPa for the base material, as reported by Fan and Cheng [19].

Table 5. Joint strengths of high-volume-fraction SiC_p/Al composites achieved by different methods.

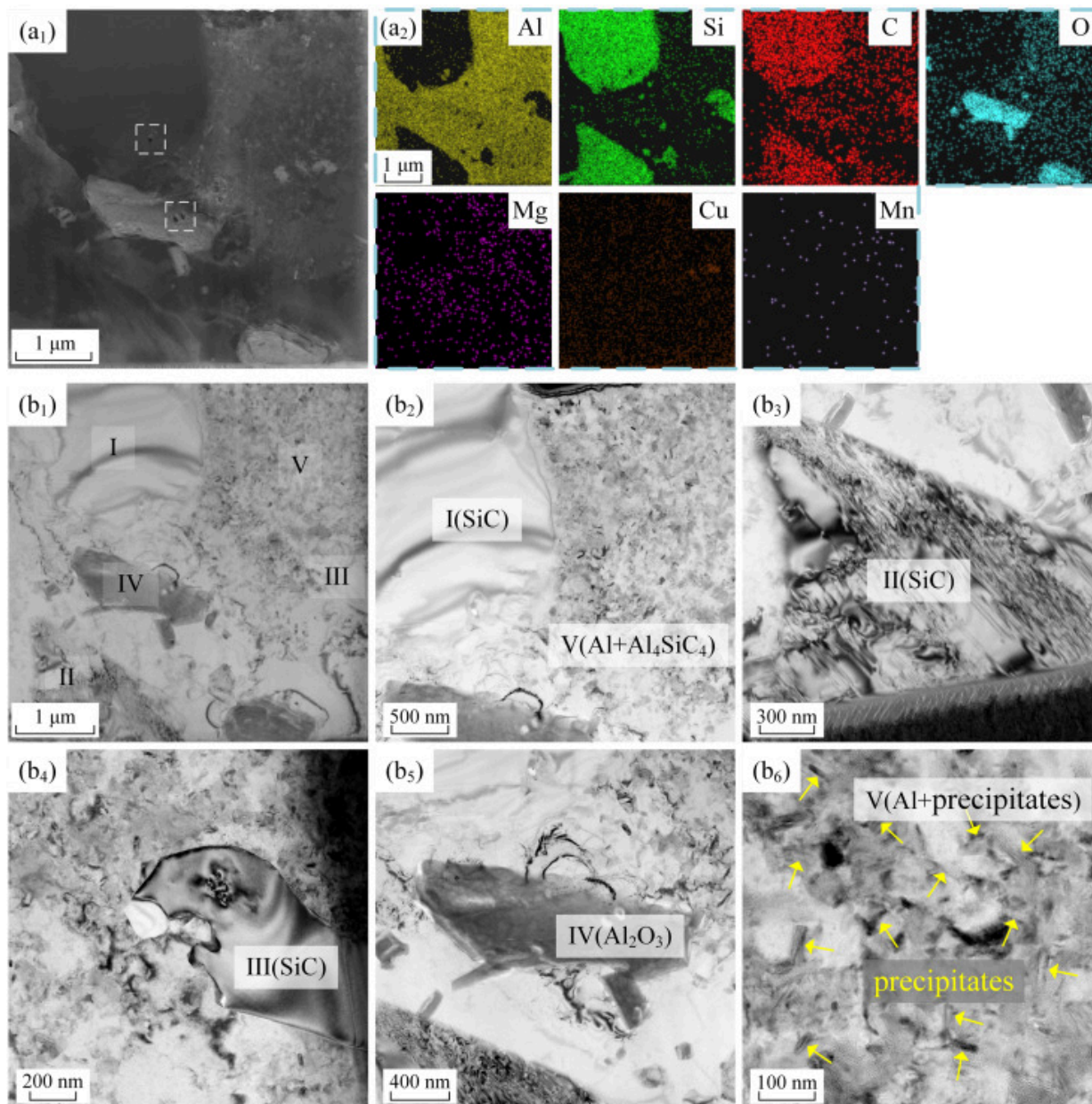
Base material	Method	Filler/Solder	Test method	Strength (MPa)	Strength ratio (%)	Reference
55% SiC _p /ZL102	Vacuum brazing	Al ₇₇ Cu ₂₀ Ti ₃ filler	Shear	75	~51	[36]
55% SiC _p /ZL102	Vacuum brazing	Al ₇₇ Cu ₂₀ Ti ₃ filler	Shear	81	~54	[19]
55 vol.% SiC _p /Al	Soldering	Sn-Ag-Cu-Al solder	Shear	52	~35	[20]
65 vol.% SiC _p /ZL102	Brazing	bismuthate glass	Shear	19.4	–	[37]

Base material	Method	Filler/Solder	Test method	Strength (MPa)	Strength ratio (%)	Reference
45 vol.% SiC _p /Al	Ultrafast laser welding	–	Bending	321	~60	Our work

3.3. The mechanism of joint formation

To clarify the bonding mechanism, TEM specimens were extracted from the Al/SiC interface within the welding seam using FIB for detailed characterization.

As shown in Fig. 9, a distinct interface is revealed between SiC and the Al matrix, free from microcracks or other defects. The voids marked in Fig. 9(a) are attributed to electron beam irradiation. Based on the BF images, elemental distribution, and previous microstructure analysis of the welding seam, the welding seam can be divided into three distinct regions, as illustrated in Fig. 9(b). Regions I-III are SiC particles exhibiting a high density of intrinsic stacking faults. TEM characterization of the base material indicated that these stacking faults are inherent to the original SiC rather than being introduced by ultrafast laser processing (Fig. A.2). Region IV is Al₂O₃, formed by oxidation during welding due to the lack of a protective atmosphere. Region V consists of an Al matrix embedded with nanoscale precipitates. These precipitates were not detectable by SEM due to their nanoscale size being below the instrument's resolution limit.

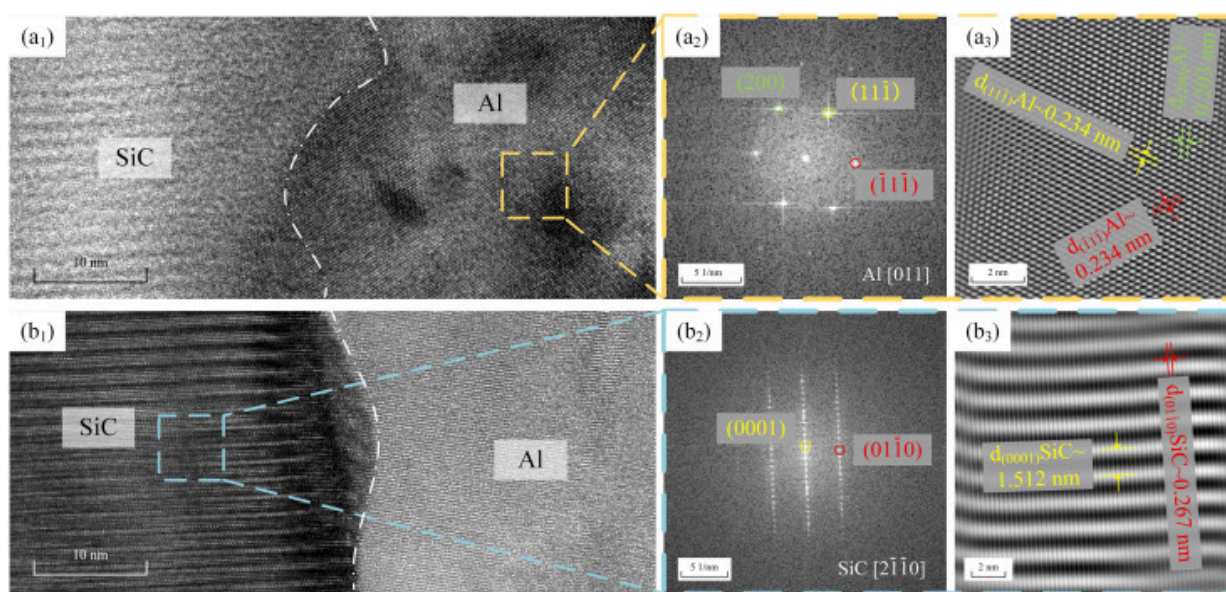


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Fig. 9. (a₁) STEM-HAADF image and (a₂) the corresponding elemental mapping of the Al/SiC interface within the welding seam. The voids marked by white rectangles are artifactual, resulting from electron beam irradiation during TEM observation; (b₁) BF TEM image of the welding seam and (b₂)–(b₆) corresponding BF TEM images of the magnified regions, respectively.

As revealed by Fig. 10, the Al/SiC interface shows no detectable interdiffusion layer or secondary phases. The interface is sharp and continuous, with the face-centered cubic Al lattice in direct contact with the hexagonal close-packed SiC lattice. The well-aligned atomic arrangement across the interface shows no signs of reaction-induced lattice distortion or discontinuity, demonstrating its atomically clean nature. In the ideal HCP structure, the (0001) reflection is systematically absent due to a zero structure factor. However, distinct (0001) diffraction spots are clearly visible in the FFT pattern from the SiC region (Fig. 10(b₂)). This deviation results from the high density of stacking faults in SiC, which disrupt the perfect ABAB stacking sequence and thereby cause the (0001) reflection to become visible.

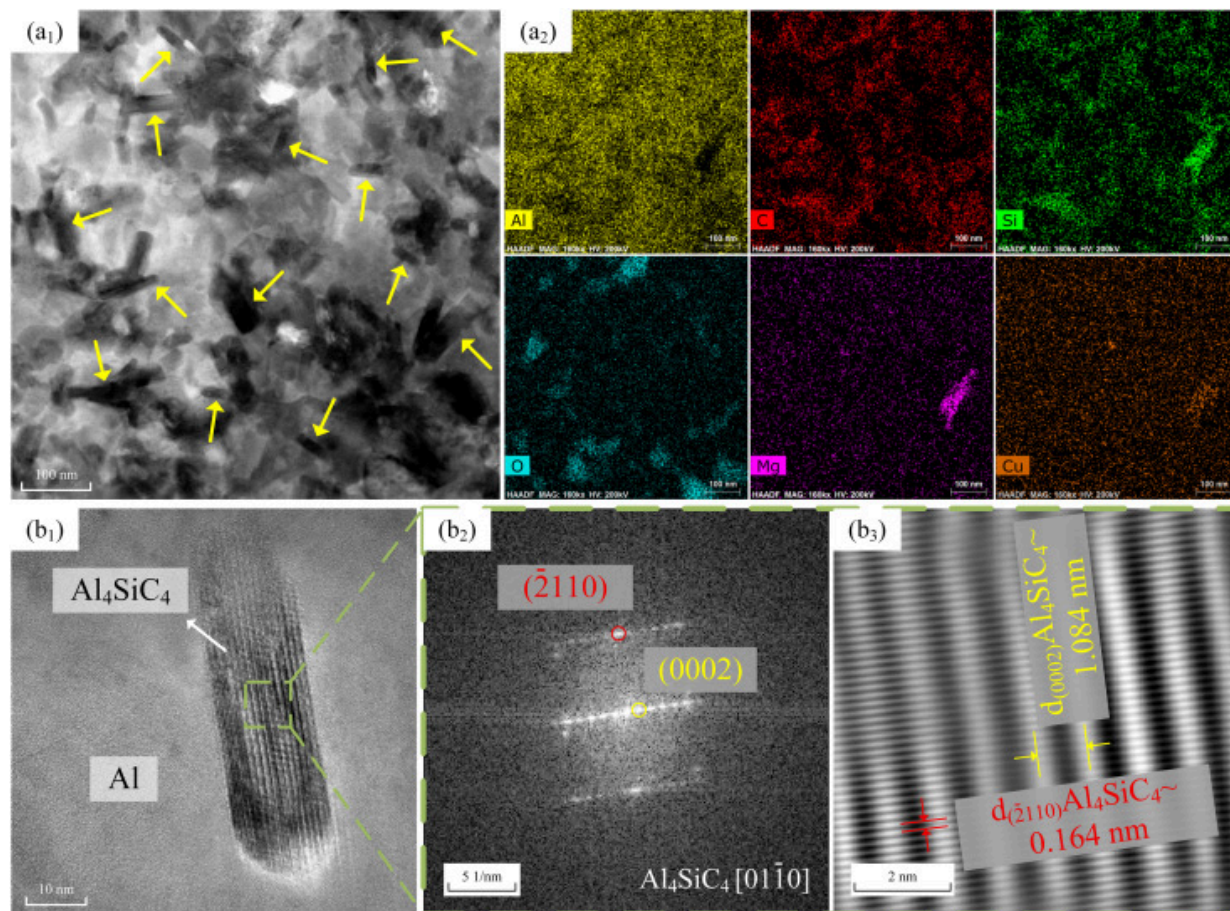


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Fig. 10. (a₁, b₁) HRTEM images acquired at the Al/SiC interface; (a₂, b₂) corresponding FFT patterns from the regions of (a₁) Al and (b₁) SiC, respectively; (a₃, b₃) corresponding IFFT filtered images.

Previous work by Pan et al. [38] indicated that in aluminum alloy systems containing both Mg and Si, the presence of Mg can promote the reaction between Al and C via the formation of Mg₂Si. In contrast, the aluminum matrix employed in this study is free of Si. Consequently, the presence of Mg is not anticipated to facilitate the formation of Al₄C₃ through the Al—C reaction. The nanoscale precipitates were identified as Al₄SiC₄ through combined elemental mapping and diffraction pattern analysis (Fig. 11). This is particularly noteworthy because in conventional fusion welding, SiC and Al typically react to form Al₄C₃, even in the absence of Si. The formation of Al₄SiC₄ observed here stands in stark contrast to the conventional fusion welding. Therefore, elucidating the ultrafast, non-equilibrium mechanisms responsible for its formation is of critical importance.

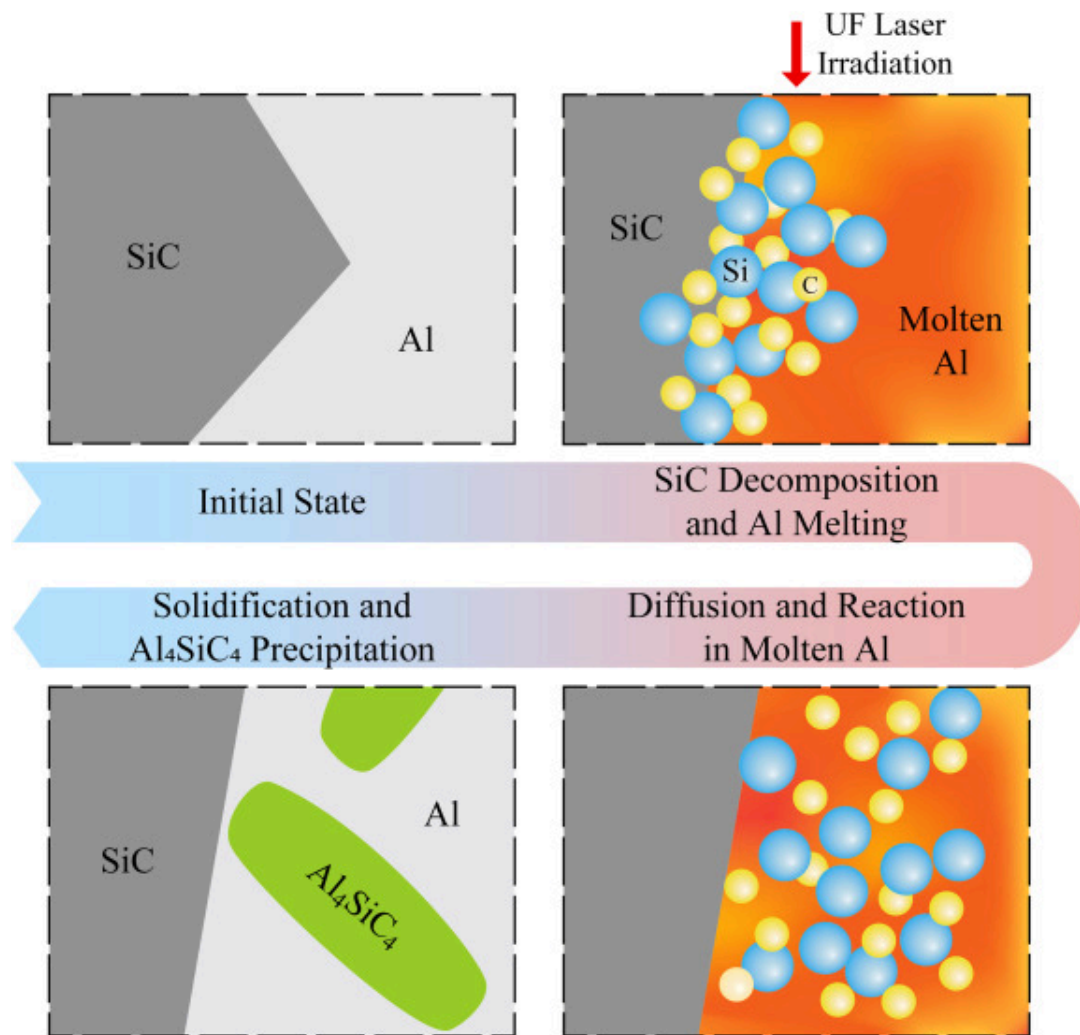


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Fig. 11. (a₁) STEM-HAADF image of the precipitate and the aluminum matrix; (a₂) elemental mapping showing the distribution of constituents; (b₁) HRTEM image of an Al₄SiC₄ precipitate; (b₂) corresponding FFT pattern from the precipitate in (b₁); (b₃) corresponding IFFT filtered image.

The formation of Al₄SiC₄ nanophases is primarily driven by the ultrafast and non-equilibrium characteristics of ultrafast laser processing. The underlying mechanism for the formation of Al₄SiC₄ nanophases is shown in Fig. 12 and summarized as follows.



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Fig. 12. Schematic illustration of the formation mechanism of Al₄SiC₄ nanophases during ultrafast laser processing.

Initially, under the extreme peak power density of the ultrafast laser, the aluminum matrix melts, while the SiC particles undergo non-equilibrium thermal decomposition

[39]. The surrounding molten aluminum lowers the kinetic energy barrier for this decomposition, thereby facilitating the decomposition reaction [40], [41]. At this stage, pores originally present in the base material evolve into bubbles within the molten pool. It is speculated that the strong Marangoni flow significantly enhances fluid stirring and mixing, thereby promoting bubble flotation and eventual escape from the melt. This mechanism contributes to the formation of a welding seam with a notably denser microstructure compared to that of the base material.

Subsequently, the liberated Si and C atoms rapidly dissolve into the melt. A plasma-induced shockwave and intense Marangoni convection promote vigorous fluid flow within the melt pool, enhancing the uniform distribution of Si and C. However, the short duration of the molten state severely limits atomic diffusion. The high volume fraction of SiC in the base material supplies abundant Si and C, leading to the formation of numerous regions enriched in these elements—referred to as compositional fluctuations.

Under these conditions, the molten pool contains an extremely high concentration of Si, which provides a sufficient Si source for the formation of Al₄SiC₄ and also effectively suppresses the formation of Al₄C₃ [42]. Hu and Baker [43] observed a significant spatial separation in the distribution of Al₄SiC₄ and Al₄C₃ within the molten pool generated by a continuous-wave CO₂ laser. More Al₄SiC₄ was found in the upper, higher-temperature region closer to the laser, while more needle-shaped Al₄C₃ was observed in the lower, cooler region. In a recent study, Xu et al. [44] employed laser oscillation to increase the peak temperature and reduce the thermal gradient within the molten pool, thereby promoting the formation of Al₄SiC₄. Their work provided a combined thermodynamic and kinetic analysis of the conditions favoring the formation of Al₄SiC₄ compared to Al₄C₃. They found that at temperatures above 1333K, the Gibbs free energy of Al₄SiC₄ is lower than that of Al₄C₃, making Al₄SiC₄ more stable at elevated temperatures. When the temperature exceeds 1550K, the critical nucleation radius r^* of Al₄SiC₄ is smaller than that of Al₄C₃, rendering Al₄SiC₄ more prone to nucleation. Transient numerical simulations by Yan et al. [39] confirm that under ultrafast laser irradiation, the lattice temperature of SiC rises dramatically within picoseconds, reaching a peak far beyond its

atmospheric boiling point of approximately 3500K. It is thus inferred that this intense heat is transferred to the molten pool, elevating its temperature and thereby promoting the formation of Al₄SiC₄.

During the subsequent rapid solidification stage, the melt, supersaturated with Si and C, provides a high driving force for the nucleation of the Al₄SiC₄ phase, which is a stable compound in the Al-Si-C system [45]. The melt attains significant undercooling, greatly increasing the nucleation rate of Al₄SiC₄ and resulting in the rapid nucleation of a high density of precipitates. The extreme non-equilibrium solidification effectively suppresses the growth and coarsening of these precipitates, restricting them to the nanoscale. Crystallographically, this phase adopts a hexagonal structure with the space group P6₃mc, exhibiting intrinsic anisotropy [46]. This anisotropy is manifested in the preferential growth along low-index crystallographic directions, such as $\langle 0001 \rangle$, with a markedly faster rate than along other orientations [44]. Consequently, the characteristic short rod-like structure forms during the subsequent rapid solidification process.

Notably, Al₄SiC₄ exhibits superior chemical stability compared to the readily hydrolyzable Al₄C₃. This enhanced stability is attributed to the strong covalent Si—C bonds in its crystal structure, which are highly resistant to attack by water molecules. Moreover, under oxidative conditions, Al₄SiC₄ forms a protective surface layer consisting of Al₂O₃ and SiO₂, which acts as a diffusion barrier against oxygen, thereby imparting excellent oxidation resistance [47], [48]. In addition, Al₄SiC₄ possesses high hardness and strength, contributing effectively to the strengthening of the aluminum matrix [49]. As it forms in situ during welding, strong chemical bonding with the matrix is achieved. The nanoscale dimensions and uniform dispersion of Al₄SiC₄ further enhance the strength of the welding seam [50]. As a result, the formation of Al₄SiC₄ significantly enhances the overall mechanical performance of the joint.

4. Conclusion

In summary, this work presents an innovative application of ultrafast laser welding for ceramic-reinforced aluminum matrix composites, offering a promising alternative for joining these otherwise challenging materials. Reliable joints of 45 vol.% SiC-reinforced aluminum matrix composites were achieved, free of macroscopic defects and exhibited a density significantly higher than that of the base material. Under optimal processing parameters, the joints reached four-point bending strengths of 321 MPa and 294 MPa in two testing configurations, which is equivalent to approximately 60% of the strength of the base material. The influence of laser power and welding speed on the joint microstructure and mechanical properties was systematically investigated. The results indicate that joint quality initially improves with increasing laser power or decreasing welding speed but degrades beyond optimal levels. Microstructural analysis revealed the formation of uniformly dispersed nanometric Al₄SiC₄ precipitates within the welding seam—a characteristic that distinguishes it from the microstructure observed in conventional laser welding. These precipitates result from the decomposition of SiC particles under ultrafast laser irradiation and subsequent in-situ reaction with the molten aluminum matrix. The Al₄SiC₄ phase exhibits superior chemical stability, and its refined, homogeneous distribution is critical for enhancing the mechanical performance of the welded joint. However, the precise absorption mechanism of ultrafast laser irradiation by SiC, along with in-situ quantification of the transient temperature, has not been fully elucidated and will therefore be a focus of our future work.

CRediT authorship contribution statement

Yina Kong: Writing – original draft, Methodology, Data curation, Conceptualization. **Bo Yang:** Validation, Supervision, Investigation. **Chun Li:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Huamin Yu:** Validation, Supervision, Resources. **Huili Cao:** Visualization, Project administration, Conceptualization. **Hao Jiang:** Validation, Supervision, Data curation, Conceptualization. **Yongxu Liu:** Visualization, Validation, Data curation. **Haolong Li:** Visualization, Investigation, Data curation. **Xiaoqing Si:** Supervision, Resources, Project administration. **Junlei Qi:** Validation,

Supervision, Resources. **Ziyang Xiu:** Supervision, Investigation, Data curation. **Jian Cao:** Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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Effects of laser power on the microstructural evolution and mechanical performance in laser dissimilar welding of TC4 to SiC_p/6092Al composite

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